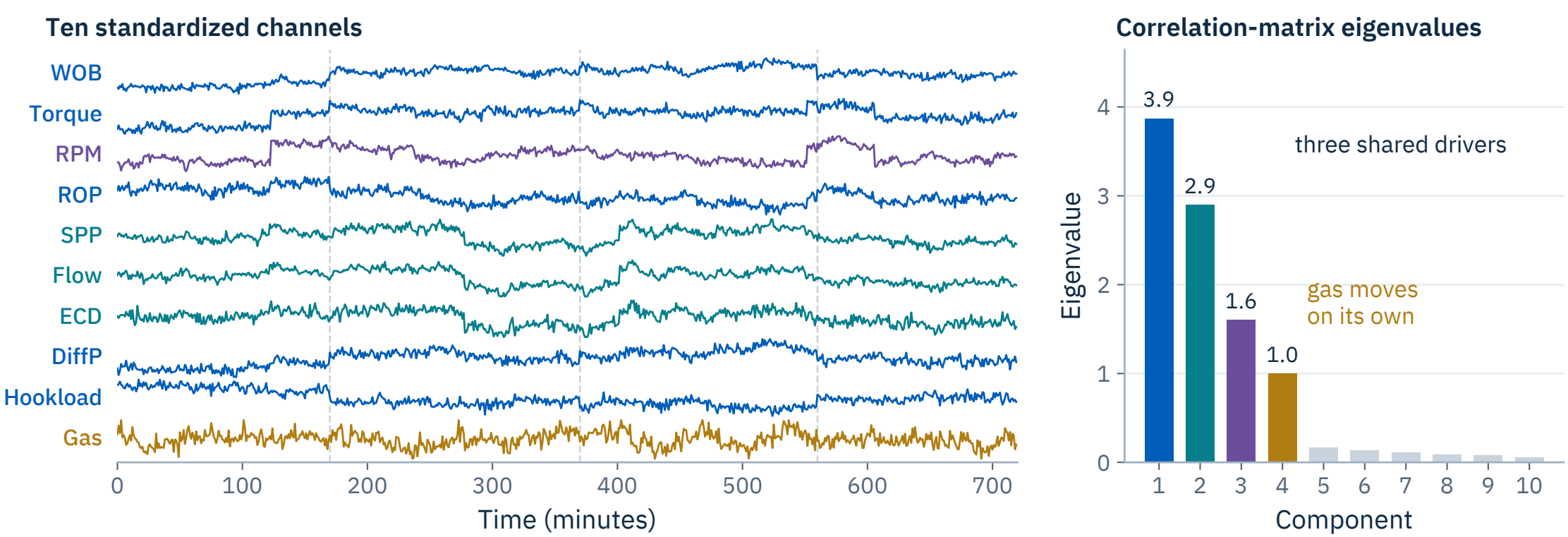

Ten standardized channels
WOB
Torque
RPM
ROP
SPP
Flow
ECD
DiffP
Hookload
Gas
0
100
200
300
400
500
600
700
Time (minutes)
Correlation-matrix eigenvalues
Eigenvalue
0
1
2
3
4
3.9
2.9
1.6
1.0
three shared drivers
gas moves
on its own
1
2
3
4
5
6
7
8
9
10
Component
Colour = hidden driver:
formation hardness
pump rate
rotary speed
none (independent)